

Emir Ceyani

Ph.D. Candidate in Electrical & Computer Engineering, University of Southern California

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Research Interests

- Large Language Models
- Graph Neural Networks
- Generative Adversarial Networks
- Flow-Based Generative Models
- Conformal Prediction
- Uncertainty Quantification
- Federated Learning
- Bayesian Methods
- AI For Science & Engineering

Education

University of Southern California , PhD in Electrical & Computer Engineering <i>Thesis: Recent Advances in GNN Training: Federation, Personalization, and Applications.</i>	Jan 2021 – Jun 2026 <i>Los Angeles, CA</i>
Bilkent University , MSc in Electrical Engineering <i>Thesis: Spatiotemporal Forecasting over Graphs with Deep Learning</i>	Jun 2018 – Dec 2020 <i>Ankara, Türkiye</i>
Bilkent University , BSc in Electrical Engineering	Sep 2013 – Jun 2018

Research Experience

University of Southern California , Graduate Research Assistant	Jan 2021 – Present
• Theory & Applications of Generative Flow Networks: Researching on the fundamentals and the applications of generative flow networks (GFlowNets) including sample efficiency, exploitation-exploration tradeoffs, and applications to hyperbolic geometry and federated learning(SIAM-SDM'25) • AI-Based Analog & RF Circuit Design: Designing graph neural networks and transformers to automate analog and RF circuit design process including topology selection, layout optimization, and AI-guided circuit design. • Conformal Prediction for LLMs: Developing uncertainty-aware LLM fine-tuning methods based on conformal prediction for close-ended and open-ended tasks to reduce the need of calibration data and extra post-hoc calibration step meanwhile improving learned representations. • Federated MRI Image Synthesis: Designed the personalized & federated image generation framework using generative adversarial networks for the first time in the literature(MIA'24, IEEE TMI'23, ISBI'23. ISMRM'23.) • Federated Graph Learning:Proposed the problem of training of graph neural networks in a federated setting and released the first benchmarking study to the literature. Improved personalization of federated graph neural networks (SIAM-SDM'25 & AAAI'22.)	
TensorOpera AI (formerly FEDML AI) , Research Scientist Intern	May 2022 – Aug 2022
• FedGraphNN: Developed a secure & private data sharing platform for graph neural networks.	
Turkcell Technologies , Research & Development Engineer	Oct 2018 - Dec 2020
• Spatiotemporal Forecasting with ConvLSTMs: Developed and deployed a spatio-temporal forecasting algorithm with convolutional LSTM models.	

Honors & Awards

- **Semi-finalist for Qualcomm Innovation Fellowship 2025.**
Abstract Title: AI Powered Revolution: Automating Analog and Radio Frequency Circuit Design
- **2025 SIAM Travel Award** for SIAM-SDM'25 conference.
- **Travel & accommodation grant** for **PRAIRE AI 2019 summer school**.
- **Travel & accommodation grant** for **Deep|Bayes 2019 summer school (Acceptance rate: $\leq 30\%$.)**
- **Funded through 5G and Beyond Joint Graduate Support Program** during M.Sc. studies by Information Technologies and Communication Authority(ICTA) (**First recipient**).
- **Full Scholarship** from Bilkent University for M.Sc. studies.
- **Research Excellence Award** by Bilkent University Electrical and Electronics Engineering Department.
- Bilkent University **High Honor student** during all semesters.
- Ranked in **top 2000** among 2M high school graduates in University Entrance Examinations(YGS/LYS), 2013.

Published and Accepted Journal Publications

- [1] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, “One model to unite them all: Personalized federated learning of multi-contrast MRI synthesis.” In *Medical Image Analysis*, Volume 94, May 2024, 103121.
- [2] G. Elmas, S. UH. Dar, Y. Korkmaz, **E. Ceyani**, B. Susam, M. Ozbey, **A.S Avestimehr**, T. Cukur. “Federated Learning of Generative Image Priors for MRI Reconstruction.” in *IEEE Transactions on Medical Imaging*, July 2023.

Publications in Conference Proceedings & Workshops (Peer Reviewed)

- [1] A. Mehradfar, X. Zhao, X., Y. Huang, , **E. Ceyani**, Y. Yang, S. Han, H. Aghasi, & S. Avestimehr (2025). FALCON: An ML Framework for Fully Automated Layout-Constrained Analog Circuit Design. Accepted to NeurIPS’25,” 2025.
- [2] **E. Ceyani**, H. Xie, B. Buyukates, C. Yang, and **S. Avestimehr**. “FedGrAINS: Personalized Subgraph Federated Learning with Adaptive Neighbor Sampling.” In *SIAM International Conference on Data Mining (SDM25)*, 2025. (oral presentation, acceptance rate: 26.7%)
- [3] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, “A Personalized Federated Learning Approach for Multi-Contrast MRI Translation,” in 31st annual meeting of International Society for Magnetic Resonance Imaging (ISMRM), Toronto, Canada, June 2023.
- [4] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, “Personalized, Federated, And Unified MRI Contrast Synthesis,” in IEEE 20th International Symposium on Biomedical Imaging (ISBI), Virtual Conference, Apr. 2023. (Presented online)
- [5] O. Dalmaz, U. Mirza, G. Elmas, M. Özbey, S. Dar, **E. Ceyani**, S. Avestimehr, and T. Çukur, “pFLSynth: Personalized Federated Learning of Image Synthesis in Multi-Contrast MRI,” in NeurIPS Medical Imaging Meets, Virtual Conference (oral), Dec. 2022. (Presented online)
- [6] C. He*, **E. Ceyani***, K. Balasubramanian*, M. Annavaram, and A.S. Avestimehr, “SpreadGNN: Serverless Multi-task Federated Learning for Graph Neural Networks,” accepted to *AAAI*, 2022. (acceptance rate of 15%) (co-first)
- [7] C. He*, K. Balasubramanian*, **E. Ceyani***, C. Yang, H. Xie, L. Sun, L. He, L. Yang, P. S. Yu, Y. Rong, P. Zhao, J. Huang, M. Annavaram, and A.S. Avestimehr, “FedGraphNN: A Federated Learning System and Benchmark for Graph Neural Networks,” in *DPML workshop at ICLR and GNNSys workshop at MLSys*, 2021.
- [8] Ergen, Tolga, and **Emir Ceyani**. ”A highly efficient recurrent neural network architecture for data regression.” 2018 26th Signal Processing and Communications Applications Conference (SIU). IEEE, 2018.

Professional Service

- **Reviewer:** ICLR’24, NeurIPS’22,’23,’25(Main & Dataset and Benchmark Tracks), AAAI’26, Federated Learning with Graph Data (FedGraph2022-2023), CrossFL-MLSYS’22, IEEE Transactions of Neural Networks and Learning Systems, IEEE Transactions of Big Data
- **Organizer & Technical Committee Member:** 1st & 2nd International Workshops on Federated Learning with Graph Data, Technical Program Committee Member at the CrossFL Workshop, MLSYS2022
- **Summer Schools Attended:** LOGML’22 Summer School, ProbAI 2021 Summer School, PAISS Summer School’19, Deep—Bayes’19 Summer School

Teaching Experience

University of Southern California, Graduate Teaching Assistant

Aug 2023 – May 2025

California, United States

- Head Teaching Assistant for EE364: Introduction to Probability & Statistics for Electrical Engineers
Fall 2024 (Instructor: Salman Avestimehr), Spring 2025 (Instructor: Michael Neely)
- Head Teaching Assistant for EE547: Applied Cloud Computing for Electrical Engineers
Fall 2023 (Instructor: Brandon Franzke)

ProbAI Summer School, Voluntary Teaching Assistant

Summer 2021

Remote

Bilkent University, Graduate Teaching Assistant

Sep 2018 – Dec 2020

Ankara, Türkiye

- Grader for EEE443/543 - Neural Networks
- Teaching Assistant for EEE 424- Digital Signal Processing
- Teaching Assistant for EEE 321 - Signals & Systems
- Teaching Assistant for EEE 493/494 - Industrial Design Project I/II

Technical Skills

Programming Languages: Python, Matlab, Java, C++, Julia

Developer Tools: $\LaTeX$, VS Code, Jupyter, Inkscape

Technologies/Frameworks: Linux, PyTorch, PyTorch Geometric, TensorFlow, NumPy, Matplotlib, Pandas, Github

Hobbies

Playing Bass, 8-ball pool, Trading card games, Martial arts, Travelling, Culinary arts